

PROJECT PROPOSAL

Title:

Real-Time Urban Sound Classification on RK3588 Using Log-Mel Spectrograms and a Lightweight CNN

Full Name:

Mingze Li

Student ID:

550374130

GitHub Username:

kolentossa

GitHub Project Link:

<https://github.com/kolentossa/elec5305-project-550374130>

GitHub Pages:

<https://kolentossa.github.io/elec5305-project-550374130>

Project Overview

Urban monitoring systems often rely on cameras or cloud services, which increase bandwidth use and can expose sensitive data. This project will develop a real-time urban sound classifier that runs locally on an RK3588 board. The system will recognise the ten UrbanSound8K classes, including sirens, car horns, drilling, engine idling, dog barking and street music. Incoming audio will be converted into log-Mel spectrograms and classified by a compact convolutional neural network. A microphone-based prototype will display the predicted class and confidence while recording only event labels and timestamps. The aim is to demonstrate accurate audio classification with bounded latency, a small model and no requirement to transmit raw audio.

Background and Motivation

Environmental sound classification supports traffic analysis, safety monitoring and context-aware devices. UrbanSound8K provides 8,732 labelled audio clips organised into ten predefined folds, enabling reproducible evaluation [1]. Earlier work showed that convolutional neural networks can learn useful patterns from time-frequency representations of environmental audio [2]. However, high-capacity models are poorly matched to continuous embedded inference. MobileNetV2 introduced depthwise convolution, inverted residual blocks and linear bottlenecks to reduce computation while preserving representation quality [3]. This project combines those ideas with audio signal processing and INT8 deployment. The topic matches my experience with computer vision, multi-object tracking and RK3588 deployment while extending that experience to a different signal domain.

Proposed Methodology

Audio will be resampled to 16 kHz mono and divided into two-second windows with a 0.5-second hop. Each window will use a 25 ms Hann window, a 10 ms frame hop and a 512-point short-time Fourier transform. Magnitudes will be projected onto 64 Mel bands and log-compressed. Per-band normalisation will be fitted only on training data. A baseline will use 20 mel-frequency cepstral coefficients with temporal mean and standard deviation features and a support vector machine. The proposed model will use depthwise-separable convolution blocks, global average pooling and a ten-class softmax layer. Clips shorter than the input window will be zero-padded, while longer clips will be segmented without mixing samples from different dataset folds. Class-balanced sampling and cross-entropy loss will be used during training. Model selection will use validation macro F1 rather than training accuracy.

Evaluation and Deployment

Training will use PyTorch with random time shifts, gain variation and time-frequency masking based on SpecAugment [4]. The official dataset folds will be preserved, and final accuracy and macro F1 will be reported across the ten held-out folds. Per-class recall and a confusion matrix will identify failure modes. The selected model will be exported through ONNX, converted with RKNN Toolkit2 and quantised to INT8 using representative training samples. Desktop FP32 and RK3588 INT8 outputs will be compared on the same test clips. Deployment tests will measure model size, P50 and P95 preprocessing-plus-inference latency, throughput and the accuracy change after quantisation. The streaming prototype will process USB microphone input and update predictions every 0.5 seconds.

Experimental Questions

The experiments will answer three questions. First, how much does the log-Mel CNN improve macro F1 over an MFCC-SVM baseline under the same fold-based evaluation? Second, which augmentation settings improve generalisation without masking class-specific acoustic cues? Third, how much accuracy is lost after INT8 conversion, and does the quantised model meet the streaming deadline on the RK3588? An ablation will compare the CNN without augmentation, with time-domain augmentation, and with combined time-domain and spectrogram masking. All model comparisons will use identical test folds and preprocessing.

Expected Outcomes

The main deliverable will be a working RK3588 prototype for ten-class streaming sound classification. A successful result will achieve macro F1 of at least 0.70, keep the INT8 accuracy loss within three percentage points of the FP32 model, and complete P95 processing in less than the 0.5-second update interval. The repository will contain preprocessing, training, conversion and deployment code; saved model files; an evaluation script; setup instructions; and a short demonstration. The final report will compare the MFCC-SVM baseline with the lightweight CNN and explain the trade-off between accuracy, latency and model size.

Timeline (Weeks 1-13)

Week	Task and Output
1-2	Confirm scope, create the public repository and reproduce the UrbanSound8K data pipeline.
3-4	Review environmental sound classification methods and implement STFT, log-Mel and MFCC features.
5	Train and evaluate the MFCC-SVM baseline.
6-8	Implement the lightweight CNN, tune training and add augmentation.
9	Run fold-based evaluation and analyse the confusion matrix.
10	Export the model, perform INT8 calibration and convert it to RKNN.
11	Deploy the streaming prototype and benchmark latency, throughput and model size.
12	Complete results, figures, README instructions and the demonstration.
13	Finalise the report, GitHub Pages site and repository release.

References

1. Salamon, J., Jacoby, C., and Bello, J. P. (2014). A Dataset and Taxonomy for Urban Sound Research. Proceedings of the 22nd ACM International Conference on Multimedia, 1041-1044. <https://doi.org/10.1145/2647868.2655045>
2. Piczak, K. J. (2015). Environmental Sound Classification with Convolutional Neural Networks. 2015 IEEE 25th International Workshop on Machine Learning for Signal Processing, 1-6. <https://doi.org/10.1109/MLSP.2015.7324337>
3. Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., and Chen, L.-C. (2018). MobileNetV2: Inverted Residuals and Linear Bottlenecks. 2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition, 4510-4520. <https://doi.org/10.1109/CVPR.2018.00474>
4. Park, D. S., Chan, W., Zhang, Y., Chiu, C.-C., Zoph, B., Cubuk, E. D., and Le, Q. V. (2019). SpecAugment: A Simple Data Augmentation Method for Automatic Speech Recognition. Interspeech 2019, 2613-2617. <https://doi.org/10.21437/Interspeech.2019-2680>